

with the degree of contribution one makes to a piece of work [35]. However, it remains unclear what constitutes *contribution* from AI, and it is generally challenging to specify which parts of the writing process make more meaningful contributions than others.

Writers worry that using AI might negatively impact their writing outcomes. In several studies, writers expressed concerns that AI might distract them from their original ways of writing, leading to lower quality of work [25]. Writers often also raised the question of whether readers would be able to tell if they were using AI to write [18], and multiple studies suggest that audiences might react negatively if they knew a piece of text was generated by AI [29, 34, 56, 68]. It remains, however, unclear whether adverse reactions result regardless of the context in which AI is being used, or whether certain use cases of AI writing assistance are more acceptable.

Writers worry that working with AI might diminish their control and joy during the writing process. In general, writers hope to maintain control throughout their writing process [11]. Writers' sense of control shapes their perceptions of work ownership, which might in turn determine whether they are using AI as a tool or being influenced by AI [39]. As such, participants of a recent study expressed strong preferences for taking AI suggestions from multiple options instead of adopting a single, complete piece of AI-generated writing [11]. From a job satisfaction perspective, writers also wondered whether working with AI would diminish the degree to which they enjoy the process of writing [18, 61].

Together, these emerging themes suggest that writers have growing concerns about AI's potential threats to authentic writing. However, **it remains unclear whether and how writers' authentic voices can be preserved when writing with AI tools.** Prior work also tends to focus only on specific aspects of authenticity. For instance, prior work studying authorship specifically addresses the *Source* aspect of authenticity, while work examining the impact of AI assistance on writing outcomes tends to conceptualize authenticity more similarly to the *Category* approach. As a result, we lack a more holistic, writer-centered view of authenticity in human-AI co-writing. Without comprehensive perspectives from writers, it remains difficult to propose solutions to regain and protect authenticity in their co-created work with AI that responds effectively to their needs and desires. Our present research aims to address these literature gaps and design goals.

2.2.2 Insights from AI-mediated communication research. Prior research on AI-mediated communication (AIMC) might bring additional perspectives on these unresolved questions around authenticity. In particular, AIMC studies have looked at how AI suggestions mediate written content for communication through text, including readers' perceptions of both the content and the message writer [22, 24, 48, 57, 59]. Importantly, writing can be viewed as a medium of communication that connects writers and their audiences, albeit often not bi-directionally or at the inter-personal level [7, 38, 52].

Indeed, various AIMC studies have shown that AI assistance (e.g., word-by-word suggestions or short-reply suggestions) can shift the characteristics of writing content, such as a more positive tone [22, 48, 59], or even affect topics addressed in writing [57]. More recently, [29] found that writing with AI assistance on opinionated content affects not only a user's writing output but also their own opinions toward the written topic. In light of these findings as well as other repeated findings around readers' negative perceptions of communicators who use AI [31, 44], scholars have expressed ethical concerns around AI's implications for information and interpersonal trust [28, 45], users' sense of agency and authenticity [59], and social relationships [24, 32, 48] as users adopt AI more widely for composing content for interpersonal communication.

While these findings suggest AI’s potential to move writers away from their own voices in writing, it is not clear whether insights from AIMC research apply to understanding AI-assisted writing more generally beyond interpersonal communication. Once again, we know little about how writers might avoid compromising their own voices when writing with AI. In particular, the connections between professional writers and their readers through writing are often different than the forms of interpersonal communication studied in AIMC, which often examines content written in mutual conversations for instrumental purposes (e.g., communicating with an owner of a rental house) or relationship-building (e.g., emailing a friend) [22, 32, 44]. Unlike such dialogues, writers seldom speak to a particular individual or aim to build interpersonal relationships with their audiences. Instead, writers use writing as a means to deliver their messages to a crowd and express themselves creatively. We thus investigate how these differences add to the complexity of authenticity in human-AI co-writing.

2.3 Designing Tools for Human-AI Co-Writing

Recent work has already proposed a wide variety of possible AI writing assistance scenarios (see our summary in Table 1), ranging from more generic to genre-specific support. Researchers have also experimented with AI assistance that supports different stages of one’s writing process as well as providing multi-form [26, 66] or multimodal [63] support at once. While many of these explorations have been shown to be helpful for writers, the majority of these studies focus on improving usability, productivity, and sometimes creativity of writing. By contrast, the possible influences of these tools on authenticity often only arise in exploratory analyses or post-study conversations with participants, leaving a noticeable literature gap.

3 METHODS

3.1 Overview of the Study Design

We examined writer-centered definitions of authenticity and the impact of AI writing assistance on authentic writing primarily through semi-structured, qualitative interviews with professional writers. To answer RQ3 and to enable participants to respond to our inquiries with situated experiences, we adopted two versions of AI-powered writing assistance tools: one with personalization through in-context learning, and one without personalization. We investigated how participants wrote with these tools in real time and delved into their co-writing experiences through interviews following each writing session. We complemented perspectives from these direct users of such emerging technology (i.e., *writers*) through a follow-up online study with indirect stakeholders (i.e., *readers*) [16]. Through these two parts of the study, we synthesize a more comprehensive view of authenticity in writing. The full study protocol (as illustrated in Figure 1) was reviewed and approved by the Institutional Review Board (IRB) of the authors’ affiliation.

3.2 Part 1: Interviews with Professional Writers

We conducted Part 1 of the study with 19 professional writers. The study includes a pre-interview survey (through an online questionnaire) where participants shared their writing samples and professional experiences as writers, and expressed their opinions on what authenticity in writing means to them and on AI writing assistance. During each interview, we first asked participants to further elaborate on their perspectives regarding authenticity and co-writing with AI. Next, participants engaged in two writing sessions to co-write a short passage with a personalized and a non-personalized AI writing assistance tool in a randomized order. Immediately after each writing session, we asked participants to reflect on their co-writing experiences through semi-structured interviews. After both writing sessions,

we asked participants to compare their experiences across the two sessions. Finally, we revealed to them the difference between the two tools and asked them a few more questions to capture their overall thoughts on co-writing with generative AI. The pre-study survey and the full interview protocol are included in Appendix A and B.

Literature	Use of language model	Types of writing	Forms of AI writing assistance	Key variables, conditions, and/or prototypes examined	Key findings
[30]	GPT-3	Argumentative writing	Text continuation/completion through word-by-word suggestions	AI suggestions prompted with positive/negative opinions of a given topic	Co-writing with an opinionated AI writing assistance shifted opinions expressed in users' writing and their attitudes toward the written subsequently.
[56]	GPT-2	Self-introduction in online profiles	Text continuation/completion through word-by-word suggestions	AI suggestions fine-tuned with specific topics	Adopting AI suggestions altered what users wrote about for self-presentation.
[5]	GPT-2	Business writing in emails	Text continuation/completion through multi-word suggestions	Length of AI suggestions	(1) Multi-word suggestions benefited ideation at the cost of efficiency in writing; (2) Non-native speakers benefited more from multi-word suggestions.
[54]	GPT-2	Short text of creative writing	Full-text generation	An interactive interface that allowed users to prompt AI writing assistance with topical and atmospheric keywords	Writers enjoyed writing with AI more than conventional writing processes and suggested AI output broadened the topics they would write about.
[60]	an RNN-based language model	Creative writing	Text continuation/completion through sentence-level suggestion	Degree of randomness (temperature) in AI suggestions	AI suggestions with lower randomness were more coherent with users' own writing, while increased randomness introduced more novel suggestions.
[17]	GPT-2	Science writing	Generating "sparks" (sentences related to a given scientific concept)	AI-generated topical sentences as inspiration for writers	(1) AI-generated "sparks" provided inspiration, translation, and perspectives to writers; (2) The quality of "sparks" did not affect users' satisfaction of writing assistance.
[36]	GPT-3	Science writing	Generating metaphors of given scientific topics	AI-generated metaphors as inspiration for writers	AI-generated metaphors offered inspiration and created positive user experiences without compromising writers' agency.
[55]	Hugging Face T5	Writing as a practice to support vocabulary learning	Generating stories based on given keywords	AI-generated stories as learning support	Users favored learning vocabularies with AI-generated stories but yielded worse learning outcomes.
[67]	GPT-3.5-turbo	Argumentative writing	Visualizing structures of written content to support writers to prototype their drafts	Visual prototyping with drafts	Visual prototyping allowed writers to experiment with and validated their ideas in drafts.
[10]	Hugging Face T5	Analytic essay	Text summarization	Real-time summarization of users' writing	Summarization helps users gain insights and strategize their own work.
[58]	LaMDA	Rewriting text into a given style	Style transfer	Effectiveness of style transfer through zero-shot learning and framing style transfer as a sentence rewriting task	The method (augmented zero-shot learning) proposed in the present work can accomplish standard style transfer tasks (e.g., changing the sentiment of text) and also be used for re-writing text metaphorically.
[26, 66]	LaMDA	Creative writing	Idea suggestion, text continuation, expansion, rephrasing, and re-writing	An co-writing interface with multiple forms of AI writing support	Designing writing support for experts need specific considerations as they have more developed writing practices and styles.
[63]	GPT-2	Creative writing	Text + image generation per users' prompt input	Benefit of providing multimodal output as sources inspiration for writers	While multimodal output could be inspiring, it also took writers more work to integrate them into their current work.
[50]	Chinchilla AI	Screenplay	Dialogue generation based on story world-building specified by writers	Effectiveness of prompting with high-level narrative arc to generate dialogues for screenplay	Writers could better focus on the overarching story building when working with this tool.

Table 1. Recent literature on human-AI collaborative writing. Here, we only include studies that examine how human users co-write with AI and the outcomes of co-writing. As such, work that studies the model/system design of AI writing assistance *without* human evaluation is beyond the scope of our literature review.